Yuyan Wang

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RESEARCH INTEREST

I am deeply interested in model deployment and decision learning within **Embodied Intelligence**. My research focuses on developing intelligent agents that can perceive, reason, and act in complex physical environments, with an emphasis on **generalization** across tasks, environments, and embodiments. I am particularly interested in **data-efficient learning** and **in-context adaptation** to enable robots to learn from limited experience and adapt to new situations with minimal additional training.

EDUCATION

University of Pennsylvania | MSE Computer and Information Science 2026.08 - 2028.05
University of Virginia | B.A. Computer Science and Statistics（GPA：3.97/4.0） 2022.08 - 2026.05

PUBLICATION

ICML 2026 Xuhui Kang, Sung-Wook Lee, Haolin Liu, **Yuyan Wang**, Yen-Ling Kuo 2025.07
Moving Out: Physically-Grounded Human-AI Collaboration
ICRA 2026 Timur Anvar, Jeffrey Chen, **Yuyan Wang**, Rohan Chandra 2025.11
Are LLMs The Way Forward? A Case Study on LLM-Guided RL for Decentralized Autonomous Driving

INTERNSHIP

Humanoid Robot Algorithm R&D Intern | BXI Robotics 2026.06 - 2026.08

- Obtain joint retargeting data for the Elf3 Humanoid Robot, train and deploy the SONIC WBC model, and achieve wholebody teleoperation and model inference for Pico VR.
- Evaluate GR00T, Pi, and LingBot-VLA, develop ELF3's task execution capabilities, and build a GR00T and SONIC control framework on ELF3.
- Conducts robot data generation research based on DreamGen, using a world model to synthesize robot task video, providing high-quality data support for improving VLA post-training and generalization capabilities.

PROJECT

Knock Until You Can See It: Cross-Modal Transfer from Sound to Vision 2026.09 - present
Supervisor: *Jiayuan Mao* | *Research Project*

- Developing an multimodal perception framework to infer latent object properties from acoustic signals.
- Investigating cross-modal knowledge transfer from acoustic observations to visual representations using VLMs.
- Optimizing active exploration and visual inference to minimize physical interactions and improve generalization to unseen objects.

NASA Lunabotics Challenge 2023.08 - 2026.05
UVA Mechatronics and Robotics Society | *CS Subteam Member*

- Achieved 1st place (Off World Grand Prize) out of over 60 universities in 2026 NASA Lunabotics Challenge.
- Use ROS2 Nav2 to explore SLAM-based localization and path planning methods, focusing on how to avoid obstacles such as craters to achieve autonomous movement of the robot on the lunar surface.
- Design action sequences by utilizing sensors and AprilTag for automated digging and dumping tasks.
- Establish a data transmission network between the control center (Linux) and the robot (NVIDIA Jetson).

Improvement and Optimization of Franka Robotic Arm Teleoperation 2024.01 - 2025.04
Supervisor: *Yen-Ling Kuo* | *Research Assistant*

- Implemented teleoperation of the Franka robot arm using a Meta Quest VR controller and space mouse for diverse manipulation tasks in both physical and Gazebo simulated environments through ROS2.
- This teleoperation setup now supports 4–6 ongoing projects in the lab across multiple research directions.